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Trouble in Paradise? A Spatial Analysis of the Camp Fire (2018)

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Introduction

This study aims to assess the current fire station infrastructure and emergency service coverage in Butte County, focusing on the 2018 Camp Fire that devastated the communities of Paradise, Concow, and Magalia and resulted in the evacuation of 40,000 people and 85 deaths. Our research question is, "How effectively did the existing fire stations in Butte County cover the population during the Camp Fire in 2018, and where should a new fire station be located using a land suitability analysis to improve wildfire emergency response?"

The initial motivation of this project was the 85 lives lost and 18,800 structures burnt, making it, at the time, the deadliest wildfire in U.S. history in over a century, which was surpassed by the Lahaina Fire in 2023 (Maranghides et al., 2021). After investigating the incident further, we were further motivated by the significant spatial challenges faced during the Camp Fire. Limited road infrastructure in and out of Paradise and the surrounding communities posed issues to responding fire personnel and evacuating civilians. Insufficient fire station coverage contributed to delays in initial response and resource requests, as well as initial attacks on the fire. The fire began on November 8, 2018, when a C-hook on PG&E's Caribou-Palermo transmission line failed, causing highly charged wires to fall into dry brush (Maranghides et al., 2021). Overgrown and poorly maintained PG&E service roads made initial containment nearly impossible for responding firefighters. Within an hour, the fire had spread over 12 miles, driven by 35 mph wind gusts and heavy fuels, and by the afternoon, over 95% of the town of Paradise had burned (Camp Fire – Fire Progression Timeline, 2021).

The decision to focus on the Camp Fire was reinforced by the recent Palisades and Eaton Fires, which further exposed weaknesses in fire response and infrastructure. At the time of our project's start, data for the Palisades and Eaton Fires were largely inaccessible since both fires were still actively burning. Realizing this, we decided to focus on the 2018 Camp Fire, which remains a critical case study for identifying and addressing these gaps we identified. The research aims to evaluate how effectively existing fire stations covered the affected population during the Camp Fire and to identify suitable locations for a new fire station using a land suitability analysis (LSA). The key objectives of this study are to identify gaps in fire station coverage that could inform future infrastructure planning. By analyzing factors such as fire station proximity, road accessibility, population vulnerability, and fire damage within the fire perimeter, this study provides insights for enhancing emergency preparedness and reducing wildfire-related risks in Butte County.

Background

According to the April 2021 National Institute of Health (NIH) report *"Wildfire impacts on education and healthcare: Paradise, California, after the Camp Fire,"* the expansion of wildland-urban interface (WUI) communities has dramatically increased over the last three decades, now encompassing one-third of US residents (Hamideh et al., 2022). In California, one out of three residents live in a WUI, meaning their homes and communities border undeveloped wildland areas highly susceptible to wildfire. Paradise's classification as a WUI community increased its vulnerability to wildfire, mainly due to its population composition. The town had many elderly residents, particularly susceptible to wildfire-related fatalities and evacuation challenges. NIH reported that 18.4% of Paradise's population was over the age of 65, compared to the national average of 16.5%, and 12.6% of the population under 65 had health issues or disabilities, higher than the national average of 8.6%. The vulnerability of this population was reflected in the Camp Fire death toll, where 66 of the 85 victims (78%) were over the age of 65, and 78 victims (91%) were over the age of 50 (Associated Press, 2019). These findings underscored the importance of considering demographic vulnerability when assessing emergency response capacity.

In addition to population vulnerability, inadequate road infrastructure and fire station coverage exacerbated emergency response challenges. Based on the California State Association of Counties classification, Paradise is considered a "Suburban" town due to its population density and settlement patterns (California State Association of Counties, 2017). NFPA guidelines require municipalities and volunteer fire departments in suburban areas with populations between 500 and 1,000 people per square mile to maintain a response time of no more than 10 minutes from dispatch to arrival on the scene (Koorsen Fire and Security, 2021). However, the California Board of Forestry and Fire Protection reported that the initial response during the Camp Fire was delayed due to limited road access and fire station proximity. Due to the challenging terrain and road network, Cal Fire personnel took approximately 30 minutes to provide the initial size-up of the fire, which was three times longer than NFPA standards. This delay in size-up directly affected the ability to request additional resources, including strike teams, water-dropping helicopters, and firefighters from surrounding counties (Rahn, 2010).

Further analysis revealed that the road network in Paradise played a critical role in the evacuation and emergency response failures. According to the National Institute of Standards and Technology (NIST) report *"A Case Study of the Camp Fire - Notification, Evacuation, Traffic, and Temporary Refuge Areas,"* the Paradise evacuation plan initially called for a staged, zone-by-zone evacuation, beginning with the eastern areas (Maranghides et al., 2023). However, the fire's rapid spread led emergency management officials to issue a town-wide evacuation order, quickly overwhelming the road network. Traffic congestion, exacerbated by morning rush hour, school buses, large vehicles, and smoke conditions, created bottlenecks on crucial evacuation routes, including Pentz Road, Wagstaff Road, Bille Road, and Pearson Road (Maranghides et al., 2023). The closure of Clark Road at 10:00 AM further compounded the issue, redirecting traffic to Skyway, which quickly became gridlocked. Burnovers occur when fire overtakes vehicles or individuals along key routes, trapping civilians and further delaying evacuation efforts. Many residents were forced to remain in Temporary Refuge Areas (TRAs) for hours until roads reopened or transportation became available (Maranghides et al., 2023). These infrastructure limitations also restricted firefighter access to burning structures and hampered their ability to suppress the fire.

The combination of population vulnerability, road infrastructure deficiencies, and fire station proximity highlighted the need for strategic improvements to emergency response infrastructure. The challenges faced during the Camp Fire informed the key focus areas of this study, identifying gaps in fire station coverage that informed our decision for a future fire station location. We hope this will enhance emergency response times, reducing the risk of future wildfire-related casualties and damage.

Methods

A variety of geospatial datasets were collected to assess fire station coverage, population vulnerability, and structural damage during the Camp Fire. The datasets included U.S. Census blocks, fire station locations, road networks, fire perimeter boundaries, and demographic data on elderly populations (see Table 1). All datasets were imported into ArcGIS Pro and projected into the same coordinate system to ensure consistency across layers. Since we were only looking at Butte County, the coordinate system used was *NAD 1983 StatePlane California II FIPS 0402 (feet)* because it was in Zone II (California Department of Conservation, n.d.). The U.S. Census block data from Esri Federal Data provided population information at the census block level, including state and county FIPS codes and decennial population counts. However, a limitation of this dataset is that it may not accurately reflect the actual population distribution at the time of the fire, reducing the accuracy of the population density. Fire station locations and road networks were acquired from OpenStreetMap. The road data included primary, secondary, and tertiary roads, which helped model evacuation routes and fire station accessibility. A notable limitation of this dataset is that it does not accurately represent real-world scenarios, such that dirt roads or unmaintained roads that would have been used during the initial fire assessment or evacuation, may not have been accounted for. The fire perimeter data was sourced from Esri, representing the 2018 Camp Fire in Paradise boundaries. The data does not account for the progression or varying intensity of the fire, which may limit the accuracy of spatial damage patterns. Structural damage data, also obtained from Esri, included the location and severity of the damage. Still, it may be incomplete since rural or isolated areas could not have been accurately reported. Demographic data on the population aged 65–69, obtained from Esri Business Analyst, served as a proxy for vulnerability. The dataset is based on an eight-year-old survey done in 2010 that may not accurately reflect the population trends in 2018.

Table 1. Datasets Used in Analysis

Datasets Name	**Data Type**	**Spatial Area**	**Attribute Fields**	**Uncertainties**	**Source**
U.S. Census Block	Vector (Polygon)	Butte County	- State FIPS Code - County FIPS Code - Decennial Population Count	It may not reflect the actual population, reducing accuracy in assessing current population vulnerability.	ESRI Federal Data
Final Fire Perimeter	Vector (polygon)	The City of Paradise	- irwinid - GISACRES	It doesn't depict the fire's progression, and reflects the variety in the intensity of the fire in different areas.	ESRI
Geofabrik - OpenStreetMaps (Roads of NorCal)	Vector (polyline)	Butte County	- fclass (primary (+ink), secondary (+link), tertiary (+link) - shape_length	Dirt or minor roads may have been used to evacuate rural residences.	OpenStreet Map
Damaged Structures	Vector (points)	Paradise Fire in Butte County	damageindicator	Buildings may not have been reported in rural or secluded areas	ESRI
Population of ages 65-69	Polygon Feature Class	Butte County	F5yearincrements_POP65C10	People moving in or out would not be accurately portrayed in an 8-year-old dataset.	Living Atlas ESRI Business
Geofabrik - OpenStreetMaps (POIS of NorCal)	Vector (point)	Butte County	fclass (firestation)	POI may not include all streets or lack updates on certain streets.	OpenStreet Map
Butte County Boundary	Vector (polygon)	Butte County	- county_ID - County_Name	N/A - was only used for visual representation of the dashboard	California Department of Fish & Wildlife

The data processing phase involved importing, cleaning, and preparing the datasets to ensure consistency and compatibility. The primary goal was to create a uniform dataset supporting network analysis, bivariate modeling, Moran's I spatial autocorrelation, generalized linear regression modeling, and suitability modeling. Before any of this, the initial step was to make sure that all datasets were projected into NAD 1983 StatePlane California II FIPS 0402 (feet) using the Project Tool to maintain spatial consistency and prevent alignment issues across layers. The U.S. Census Block data from the ArcGIS Hub provided the foundation for the population, which was necessary for analyzing the population coverage of existing fire stations. The Select by Attribute tool was used to filter the state FIPS code to 06 (California) and the county FIPS code 007 (Butte County) to create a feature class representing only the census blocks within Butte County. After the selection, the dataset was exported using the Export Feature Class tool to create a clean, focused layer. Following the census data import, we imported the fire station data and road network using a shapefile downloaded from OpenStreetMap's Geofabrik portal. The Select by Attribute tool was used to isolate fire station data from the broader points of interest dataset. Once the fire stations were identified, the Select by Location tool was applied with an intersect relationship to refine the selection of fire stations within Butte County. This ensured that only the fire stations relevant to the study area were included in the analysis. The roads were also sourced from the OpenStreetMap's Geofabrik portal, where we used the Select by Attribute tool to isolate roads classified as primary, primary link, secondary, secondary link, tertiary, and tertiary link (OpenStreetMap contributors, 2025). The selected road network data was then exported to create a refined dataset as a new feature class. Finally, the Select by Location tool was used with an intersect relationship to extract road segments within the Butte County Census Blocks, ensuring that only roads within the study area were retained for analysis. Structural damage data posed more challenges due to its format and lack of editability. The damaged data was stored as part of a geodatabase (GDB), which prevented direct editing or field calculations. To address this, the dataset was exported using the Feature to JSON tool, which converts feature layers into JSON format (JavaScript Object Notation). The output was set to GeoJSON format, creating a new editable dataset. The new JSON file was then processed using the JSON to Feature tool, which converted the GeoJSON into a feature class representing damaged structures. With an editable feature layer created, we added a new field called "damaged_lv." This field allowed us to assign numerical values to the damage categories previously stored as text values. Using the Calculate Field tool with Python3, we converted the qualitative description, for example, the “Destroyed (>50%)”, to a quantitative value of “4”. For improved visual representation, we created a kernel density map, which produced a heat map showing the concentration of structural damage within the Camp Fire perimeter. This visualization provided insight into which areas experienced the most severe damage and informed the later stages of the suitability analysis. From these processes, the necessary datasets were now within Butte County and had compatible coordinate systems for future processes.

The first analysis method we used was a Network Service Area Analysis to assess the fire station coverage and response capability during the 2018 Camp Fire (Figure 1). The goal was to evaluate the driving time from each fire station and determine the population coverage within different response time intervals. The driving time cutoff values were set to 5, 10, and 15 minutes to reflect different levels of response capability. The output type was set to polygons to create a clear visual representation of coverage. The geometry was set at overlaps to ensure accurate spatial representation, allowing areas of overlapping coverage from multiple stations to be shown. The output geometry was set to dissolve and rings to display distinct bands for each driving time interval.

After running the service area analysis, the next step was creating a model to compute and extract the results for each individual time interval (Figure 2). This model allowed for an automated analysis of fire station coverage at 5, 10, and 15-minute driving times across Butte County. The service area output provided insights into existing coverage gaps and areas where fire station response times could be improved. This led us to our future analyses since the underserved population needed improvements.

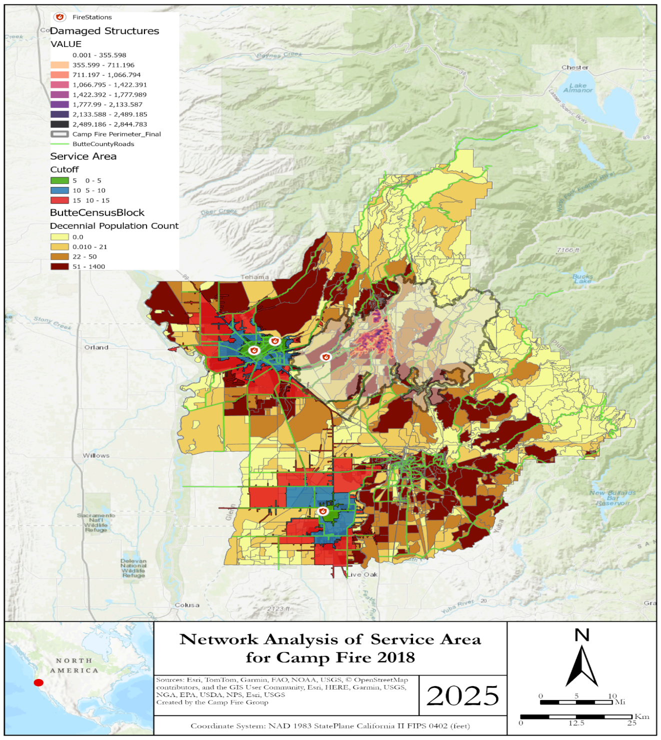

Figure 1. Network Analysis of Butte County

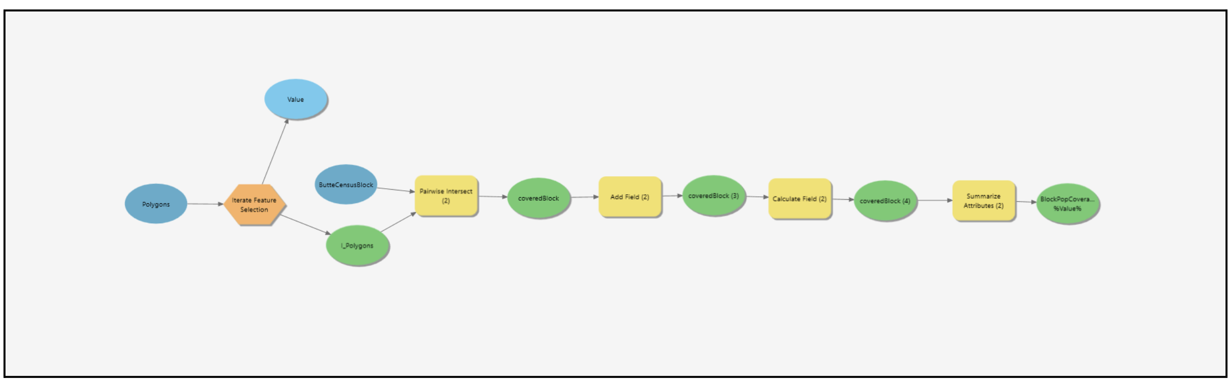

Figure 2. Model Used for Network Analysis Output Tables

The next step was to run a bivariate analysis and a generalized linear regression model. First, we needed to integrate all data layers into a single dataset with a consistent spatial resolution, coordinate system, and projection. This process allowed us to assess localized and countywide patterns in fire station coverage and structural damage, helping inform future placements. The census block layer, which had the smallest spatial resolution and encompassed the entire study area (Butte County), was chosen as the foundation for integrating the other data layers. This decision allowed us to aggregate and normalize data at a consistent scale. The first layer added to the census blocks was the road network data from OpenStreetMap. We used the Summarize Within tool to calculate the total length of roads within each census block. The summary field was set to shape length, and the sum was computed in kilometers. This provided a measure of road network density, which was expected to influence fire station accessibility and response times. Next, we incorporated the structural damage data. Although the kernel density layer created earlier provided a useful visual representation of damage, it lacked an attribute table necessary for numeric analysis. Therefore, we used the structural damage dataset that had been calculated to have the average of the structural damage per census block. To simplify the analysis, we created a new field called "damageindicator". We used the Calculate Field tool with a Python function to convert text-based damage values into numeric values: “1 if !mean_damage_lv! > 0 else 0”. This created a binary classification: one represented census blocks with structural damage, and zero represented undamaged blocks. We again used the Summarize Within tool to compute the sum of structural damage within each census block, adding the damage data to the same layer that already contained road density information. After establishing road network and structural damage data within the census block layer, we added demographic data on the population aged 65 to 69 using data from Esri Business Analyst. This dataset was chosen based on evidence that seniors were disproportionately affected during the Camp Fire evacuation. The data was joined using a spatial join to the existing census block layer. Next, we calculated the distance from each census block to the nearest fire station. We used the Feature to Point tool to generate centroid points from the census blocks, followed by the Near Tool to compute the shortest distance from each point to the nearest fire station within the final fire perimeter. After computing the distances, we performed a Spatial Join to integrate the distance values into the census block polygon layer. The final step was using the Intersect Tool to limit the dataset to areas within the final fire perimeter and refine the study area to only the fire-affected zones. This ensured that areas unaffected by the fire would not distort our future analyses. The combined dataset now contained population density, road network density, distance from the fire station, structural damage, and vulnerable population metrics, all within the final fire perimeter. We ran a bivariate (Figure 3) and a Moran’s I spatial autocorrelation analysis (Figure 4). A bivariate local analysis (BLA) allows for examining localized spatial relationships between the fire station proximity and structural damage to determine whether proximity influenced fire damage patterns at a neighborhood scale. Moran's I spatial autocorrelation evaluated whether the residuals from the analysis exhibited clustering or random distribution, which helps identify additional spatial factors, such as terrain or wind patterns, that could be other contributors to structural damage.

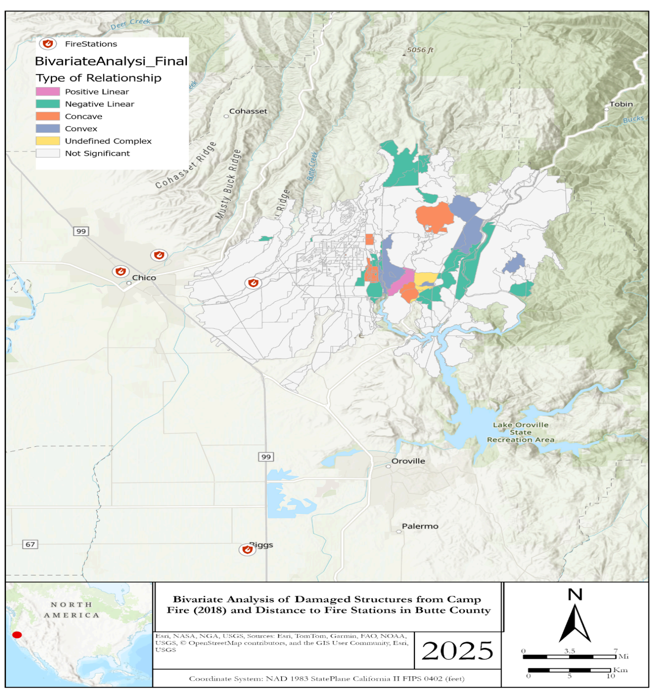

Figure 3. Bivariate Analysis

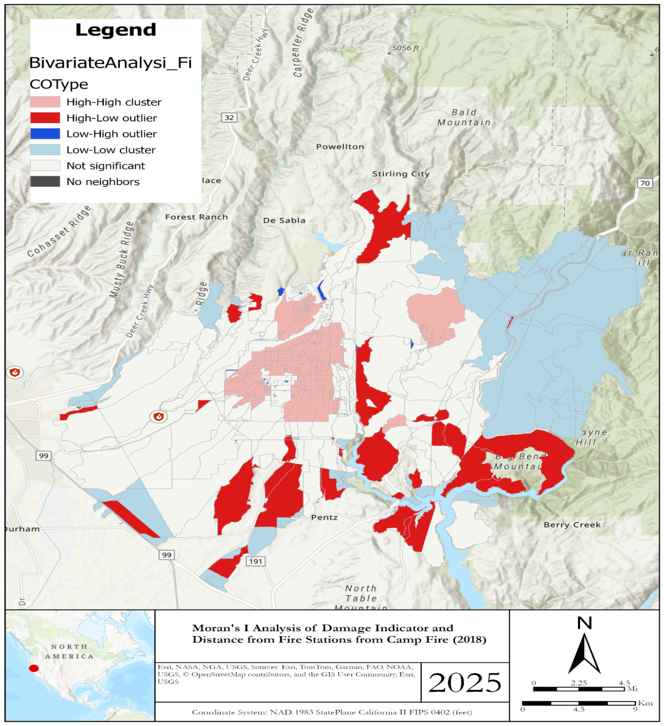

Figure 4. Moran's I Analysis of Camp Fire

The next step in our analysis plan was to conduct a generalized linear regression (GLR) with the prepared integrated dataset (see Figure 5). The GLR was conducted to assess the relationship between a dependent variable (e.g., fire damage) and multiple explanatory variables (e.g., population density, road network length, distance from fire stations, and elderly population) while accounting for non-normal data distributions. The GLR allowed for the evaluation of both the strength and significance of these relationships, helping to identify which factors significantly influence the outcome and providing a predictive framework for informing future infrastructure decisions, such as optimal fire station placement.

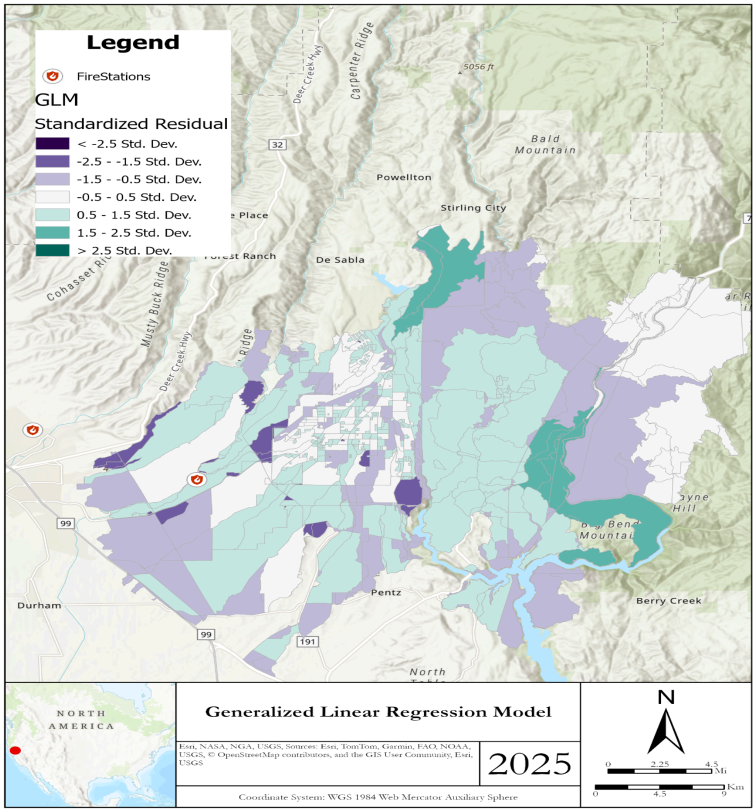

Figure 5. Generalized Linear Regression Model

The final step involved conducting a land suitability analysis using the Suitability Modeler to identify the optimal location for a new fire station in Butte County. The explanatory variables and their respective weights were based on the coefficients obtained from the GLR model, ensuring that the suitability analysis reflected the statistical relationships uncovered during the regression analysis. To prepare the data for the suitability analysis, all individual vector layers were converted to raster format using the Feature to Raster tool. This step ensured that all layers shared a consistent spatial resolution and extent necessary for an accurate multi-criteria analysis. The raster layers were then imported into the Suitability Modeler, where each was assigned a continuous function and a weight based on its significance in the GLR model. The “Feature_Pop” layer represented the decennial population and was weighted at 15%. While population density is significant in assessing structural damage, it is not as influential in determining future emergency response locations as other factors, such as accessibility and vulnerable population density. The continuous function was set to linear because the relationship between population density and suitability for a fire station is expected to increase steadily without sudden jumps. The “Feature_AGE” layer represented the population aged 65 to 69, a key demographic identified as vulnerable during the Camp Fire. The weight was set to 25% because older populations faced higher evacuation challenges and increased risk during emergencies, making it a critical factor in planning future emergency response infrastructure. The continuous function was set to linear because the risk associated with vulnerable populations is expected to increase population density in the affected age group gradually. The “Feature_DIST” layer was assigned the largest weight of 40% because it reflected the distance to the nearest fire station, which was a massive inhibitor during this event. A linear function was used because the relationship between response time and distance is expected to be relatively consistent because greater distances predictably result in longer response times. The “Feature_ROAD” layer represented the road network density and was assigned 20% because road accessibility was a considerable limiting factor during the Camp Fire. Limited road access restricted evacuation and emergency response, exacerbating the damage. A linear function was used because the impact of road availability is expected to increase consistently with greater road density.

As stated above, a linear function was selected for all criteria because the relationship between the suitability score and the underlying variable is expected to increase or decrease consistently without any non-linear thresholds or abrupt changes. As road network density increases, accessibility and response times improve consistently, making a linear function the most straightforward and accurate model to describe this relationship. Additionally, using linear functions allows for smooth transitions in the suitability surface, preventing artificial clustering or abrupt changes in suitability zones. The final suitability model output was a raster dataset, with each cell assigned a suitability score based on the weighted sum of the contributing factors (Figure 6). This map provided a clear visual representation of the most favorable locations for future fire station placement, helping to inform strategic emergency response planning.

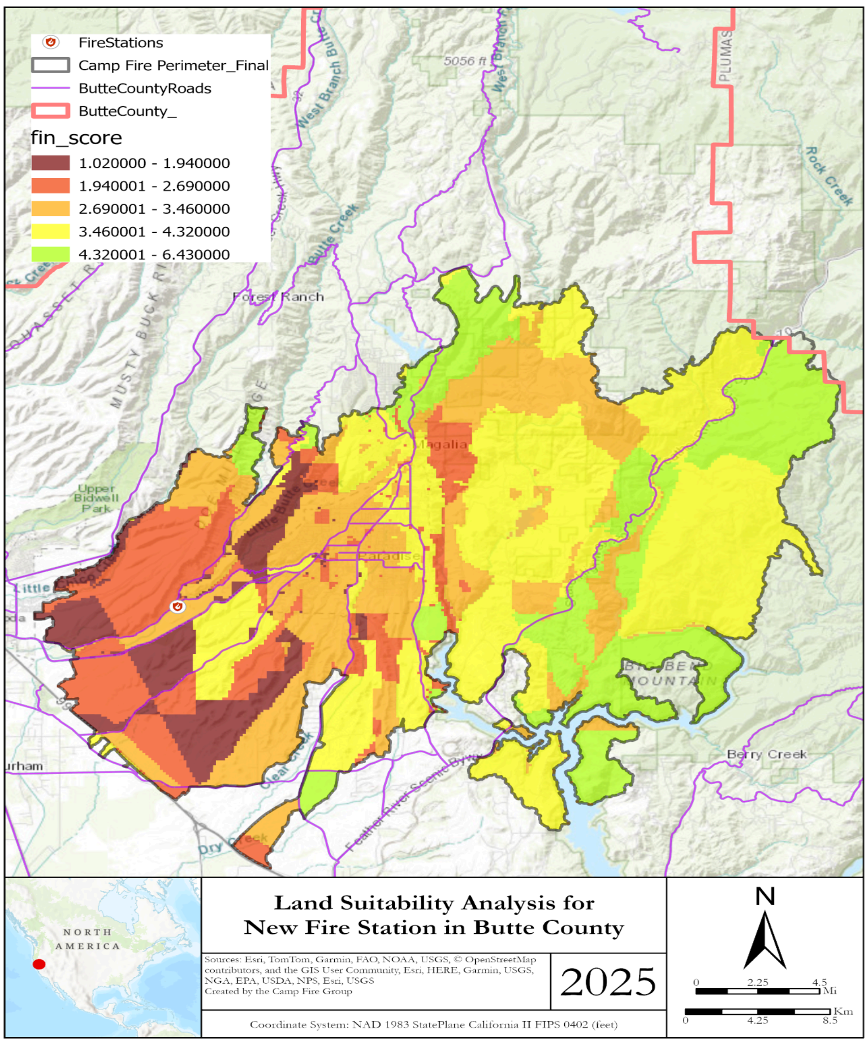

Figure 6. Final Land Suitability Analysis for New Fire Station

Application Results

Based on our methodology, our analysis employed bivariate local analysis, Moran's I spatial autocorrelation, and Generalized Linear Regression (GLR) to assess the spatial relationships between fire station coverage and structural damage during the 2018 Camp Fire. This multi-modeling approach allowed us to evaluate localized and broader fire response efficiency and vulnerability trends, contributing to a comprehensive understanding of emergency service gaps supported by statistical data.

The bivariate local analysis assessed the localized relationship between fire station distance and structural damage at the neighborhood level. As shown in Figure 3, the bivariate local analysis model revealed mixed and largely non-significant relationships across the fire perimeter. The categorical summary shows that only seven blocks (0.93%) exhibited a significant positive linear relationship, indicating that shorter distances to fire stations were associated with less structural damage. Conversely, 33 blocks (4.41%) displayed a negative linear relationship, suggesting that increased distance correlated with more significant damage. However, most features (87.05%) were insignificant, indicating that fire station proximity alone was not a strong predictor of damage within the final fire perimeter. This limited significance suggests that other spatial factors, such as fuel (vegetation), steep terrain, wind, or fire behavior, likely influenced the variability in structural damage more than fire station proximity alone. Additionally, concave and convex patterns accounted for 3.47% and 2.27% of the results, respectively, indicating non-linear effects that could reflect the influence of geographic or infrastructural barriers (Figure 3).

Given the limited explanatory power of the bivariate local analysis model, we expanded our scope using a generalized linear regression to assess the trends within the final fire perimeter. The generalized linear regression incorporated multiple explanatory variables, including population density (POP100), road length (SUM_LENGTH_KILOMETERS), elderly population percentage (F5YEARINCREMENTS_POP65C10), and fire station proximity (NEAR_DIST). The generalized linear regression results showed that road length and the elderly population had the most statistically significant impact on structural damage. The GLR model produced an adjusted R-squared value of 0.243640, indicating that the selected variables could explain approximately 24.36% of the variation in structural damage. Key findings from the GLR model include population density (POP100), which had a coefficient of -0.000952 ($p = 0.024870$), indicating that higher population density slightly decreased the likelihood of structural damage. This suggests that denser areas may have had better emergency response access or fire-resistant infrastructure. Road length (SUM_LENGTH_KILOMETERS) had a coefficient of -0.045160 ($p = 0.031150$), suggesting that more extended road networks were negatively correlated with structural damage. This could reflect improved evacuation routes or firefighting access. The elderly population percentage (F5YEARINCREMENTS_POP65C10) had a coefficient of 0.029961 ($p = 0.00000$), indicating that higher percentages of elderly residents increased vulnerability to fire damage. This aligns with findings from previous wildfire studies highlighting the evacuation challenges older populations face. Distance to the nearest fire station (NEAR_DIST) had a coefficient of -0.024373 ($p = 0.00000$), suggesting that increased distance to fire stations correlated with higher structural damage. However, the relatively small coefficient magnitude indicates that fire station proximity alone was not the dominant factor influencing fire impact.

Next, we conducted a Global Moran's I analysis to assess spatial clustering in the generalized linear regression residuals (Figure 4). The Moran's I Index was 0.521 with a Z-score of 30.59 and a highly significant p-value of 0.000, confirming that the residuals were strongly clustered rather than randomly distributed. This indicates that fire damage was influenced by spatially structured processes that were not fully captured by the generalized linear regression model. Specifically, clustering high-high outliers (red areas in Figure 4) around the town of Paradise highlights localized fire impact patterns, which may reflect topography, wind patterns, and infrastructure deficiencies.

The land suitability analysis integrated the findings from the generalized linear regression to determine the most effective locations for a new fire station in Butte County. The land suitability analysis identifies the most suitable areas based on factors such as fire station proximity, road density, population density, and age vulnerability. The final suitability map (Figure 6) reveals the range of suitability scores, with the highest-priority areas in green and the lowest-priority regions in dark red. Higher suitability scores reflect areas with better access to infrastructure and high-risk populations, while lower scores indicate poor coverage and challenging terrain. The map highlights that the central and northeastern regions of the county, particularly near Paradise and Magalia, are the most suitable locations for a new fire station, where clusters of high damage coincide with gaps in service coverage. This alignment reflects the findings from Moran's I analysis and the generalized linear regression, which indicated that proximity to fire stations was not the sole predictor of structural damage. High-priority areas are also positioned along major roadways, reinforcing the importance of road network accessibility for effective emergency response. Conversely, the western and the southwestern regions show lower suitability scores due to poor road access and existing coverage gaps, presenting logistical challenges for emergency response. The map underscores the strategic need for improved fire station infrastructure in the eastern and southeastern parts of the fire perimeter, where road access is limited and damage is concentrated. For an interactive experience, refer to the dashboard created on Experience Builder through ArcGIS Online: https://experience.arcgis.com/experience/6b4f1ac982e94c0a924fd639dc29eb53/.

Discussion

The initial network analysis conducted to assess the coverage of fire stations within Butte County led to the conclusion that the fire coverage provided was highly insufficient. With only a tiny portion of the total area of Butte County being covered within the optimal 10-minute service time window, it became apparent that the current number of fire stations was incapable of meeting demand. Furthermore, when the service coverage network analysis is overlaid with the final Camp Fire perimeter, only the edges of the perimeter receive optimal coverage. Therefore, this first network analysis served as our foundational map for the project, proved that another fire station was needed, and strengthened our research claim.

Setting decennial population density, road network density, elderly population (65-69), and distance to the nearest fire station as our explanatory variables, we ran a generalized linear regression model where the response variable is damaged structures. We conducted GLR to observe whether these variables had a significant relationship with damaged structures. Regarding our explanatory variables, road density and fire station proximity were substantial predictors of fire-damaged structures while having negative coefficients. The negative coefficients indicate that higher road density and closer proximity to the fire stations correlate with reduced structural damage. Elderly populations (65-69) displayed significance with a positive coefficient, highlighting that the elderly were positively associated with structural damage and that their populations were among the most vulnerable during the Camp Fire. In comparison, decennial population density displayed significance and a negative coefficient, suggesting that more densely populated areas in Butte County tended to experience less structural damage.

We encountered several data limitations through different stages of our project. There was a lack of available humidity, wind speed, and fire spread data compatible with our project due to the resolution not being small enough to be mapped in Butte County. The compatibility issues led to our team deciding to exclude these variables, inherently limiting the project's ability to explain why some areas experienced more damage despite station proximity. Furthermore, the federal datasets utilized for analysis are slightly dated, with updates to some of the data not being made in several years. Most notably, the ESRI population data for 65-69 was last updated in 2010, eight years before the Camp Fire. Additionally, according to our literature review, this dataset did not encompass all ages we believe are most vulnerable.

The Business Analyst tool provided this population of ages 65-69. However, we wished to have included ages 65+, as the mean was 72, and someone was even 99. Our land suitability analysis was confined to the final fire perimeter. This meant that the model could not account for areas divided by water. The project was limited by the capabilities of the model, therefore, assigning areas that faced significant geographic barriers higher suitability scores. Current data and land suitability models cannot represent the real-world geographic limitations on where a new fire station would be most suitable.

Further research should be conducted to analyze the relationships between different criteria on fire impact and reactivity. Researching the suitability of new roads in Butte County would broaden the potential locations suitable for fire stations in Butte County. Such road network research could lead to drastically different network analysis results that could improve the response time of firefighters, as they would gain faster access to previously inaccessible areas where there could be fires. Similarly, expanding the age groups analyzed as vulnerable populations to account for youth and young families would allow us to explore the assumption that they experience higher vulnerability during fires than adults. The assumption is that socioeconomic levels would be a significant variable for determining a new fire station and exploring whether lower-income communities received adequate fire response. Instead of researching the human influences on fire impact and destruction in communities, further research studying the flammability of vegetation based on explanatory variables such as height, moisture, and type could be used to observe their impact on fire spread and intensity in Butte County.

Conclusions

This study evaluated the effectiveness of existing fire station coverage in Butte County during the 2018 Camp Fire and identified optimal locations for a new fire station using a land suitability analysis. The generalized linear regression results revealed that distance to fire stations, road density, and elderly population were significant predictors of fire damage, with fire station proximity showing a relatively small effect. The Moran's I analysis confirmed that residuals were strongly clustered, indicating that additional spatial factors, such as terrain and wind patterns, likely influenced damage patterns. The LSA highlighted high-priority areas near Paradise and Magalia, where clusters of high damage coincided with gaps in fire station coverage and limited road access. These findings underscore the importance of improving road infrastructure and strategically placing fire stations to enhance emergency response capacity and reduce future wildfire vulnerability in Butte County.

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